



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

BDCTW401

TILE WORKS

Perform tile works

Competence

RQF Level: 4

Learning Hours



70

Credits:7

Sector: Construction and Building Services

Trade: Building Construction

Module Type: Specific

Curriculum: CBSBDC4001- TVET Certificate 4 in Building Construction

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Purpose statement	This module describes the knowledge, skills, and attitude to be acquired in order to place well tiles on the walls, stairs and floors. It is intended to be pursued by learner at TVET level IV Building Construction. At the end of this module, the learner will be able to prepare working area, select materials, tools and equipment. He/She will also be able to cut, apply tiles, and perform grouting and cleaning of the workplace.					
Leaning assumed to be in place	Concrete works, Building construction drawing and safety at workplace.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Prepare the work	1.1. Construction drawing is correctly interpreted based on information required.
	1.2. Tools are appropriately prepared according to the work to be done.
	1.3. Equipment are appropriately prepared in accordance with their function.
	1.4. Workplace is properly prepared according to the work to be done.
	1.5. Materials are properly prepared according to the work to be done.
2. Prepare mortar	2.1. Mixing place is adequately prepared as per standard requirements.
	2.2. Mortar materials are properly collected according to their types.
	2.3. Mortar materials are accurately batched in accordance with mix ratio.
	2.4. Mortar materials are correctly mixed according to the mixing methods.
3. Lay tiles	3.1. Area to be tiled is accurately set out in accordance with designed levels.

	3.2 . Tiles are accurately cut according to the place to be used.
	3.3 . Tiles are properly applied to the surface according to the designed pattern.
	3.4 . Tiles joints are adequately grouted according to the specification.
4. Clean the workplace	4.1. Tiled surface is properly cleaned after work.
	4.2. Working area is properly cleaned after work.
	4.3. Used tools and equipment are properly cleaned according to their type.
	4.4. Used tools and equipment are correctly stored according to their category and status.

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Identify tools and equipment used for tile works ✓ Identity different types of tiles ✓ Identify different types of tile adhesives	✓ Interprets construction drawings ✓ Draft schedule of work ✓ Handle tile works 'Tools, Equipment and material ✓ Lay different types of tiles ✓ Apply different types of adhesives ✓ Apply different types of tiles patterns ✓ Take measurements ✓ Estimate quantity of materials ✓ Cut tiles	✓ Demonstrate awareness of workplace hazards and associated control measures ✓ Demonstrate persistent, team working, flexibility, punctuality at work ✓ Demonstrate the ability to Work independently with minimal supervision ✓ Respect time assigned to every task ✓ Show innovation and creativity for tile patterns design and tile laying

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Prepare the work 2. Prepare mortar 3. Lay tiles 4. Clean the workplace
Learning outcome 1: Prepare the work	Learning hours: 5
Indicative content	
• Identification of tools used in tile works ➤ Batching tools ➤ Cutting tools ➤ Transporting tools ➤ Mixing tools ➤ Tracing tools • Use of tools used in tile works • Identification of equipment used in tile works ➤ Safety equipment ➤ Tile cutter ➤ PPE • Use of tools used in tile works • Identification of materials used in tile works ➤ Tiles ➤ Cement ➤ Grouting materials ➤ Spacers ➤ White cement ➤ Sand ➤ Water ➤ Additive • Qualities of materials used in tile works • Workplace preparation ➤ Tools used in cleaning workplace ➤ Methods of cleaning workplace	
Resources required for the learning outcome	
Equipment	– PPEs, Safety equipment, wheelbarrow, electric tile cutter.
Materials	Ceramic tiles, Mason line, Spacers, Grouting materials, Cement, Adhesive materials, tile cutter blade,
Tools	Tape measure, decameter, tile plier, tile sliding cutter, tile wet cutter, tile cutting pen, hand grinder, rotary grinder, tile trowel, tile string, mallet hammer, magnetic level. Plastic hammer.
Facilitation techniques	• Group discussion

	• Trainer guided • Brainstorming • Trainer guided
Formative assessment methods	• Open ended questions • Matching types questions • Multiple choice questions

Learning outcome 2: Prepare mortar		Learning hours: 15
Indicative content		
• Types of mortar ➤ Cement mortar ➤ Lime mortar ➤ Cement –lime mortar ➤ Mud mortar • Identification of materials used for mortar • Process of mortar preparation ➤ Selection of tools, equipment and materials ➤ Preparation of work place ➤ Batching of ingredients ✓ Volume batching ✓ Weight batching ➤ Mixing of ingredients ✓ Hand mixing ✓ Machine mixing		
Resources required for the indicative content		
Equipment	PPEs, Safety equipment, wheelbarrow, mortar mixer	
Materials	Cement, lime, water, sand, additives	
Tools	Gauge box, spades, buckets, pans	
Facilitation techniques	• Group discussion • Trainer guided • Brainstorming • Trainer guided	
Formative assessment methods	• Open ended questions • Matching types questions • Multiple choice questions • Practical exercises	

Learning outcome 3: Lay tiles		Learning hours: 45
Indicative content		
• Types of tiles • Process of laying tiles ➤ Set out the area ✓ Identification of tools and equipment ✓ Marking of the centre line ✓ Marking of the midpoint of the room ✓ Allocation of the starting point ➤ Apply tiles on structure elements ➤ Grouting of tiles Joints ✓ Permeation grouting ✓ Compaction grouting ✓ Vacuum grouting ➤ Cleaning tiled surface ✓ Foaming range ✓ Spraying range ✓ Alcohol is flashing		
Resources required for the indicative content		
Equipment	– PPEs, Safety equipment, wheelbarrow, electric tile cutter,	
Materials	Ceramic tiles, Mason line, Spacers, Grouting materials, Cement, Adhesive materials, tile cutter blade,	
Tools	Tape measure, decameter, tile plier, tile sliding cutter, tile wet cutter, tile cutting pen, hand grinder, rotary grinder, tile trowel, tile string, mallet hammer, magnetic level.	
Facilitation techniques	• Group discussion • Trainer guided • Demonstration • Brainstorming	
Formative assessment methods	• Open ended questions • Matching types questions • Multiple choice questions • Practical exercises	

Learning outcome 4: Clean the workplace		Learning hours:5
Indicative content		
• The purposes of cleaning tools and equipment • Cleaning tools and equipment ➤ By water ➤ By oil ➤ By brush ➤ By air compressor • Process of storing tools and equipment		
Resources required for the indicative content		
Equipment	Air compressor, PPEs	
Materials	Water, oil, soap	
Tools	Broom, hand broom, bucket, floor scrubbing, brushes, vacuum cleaner, duster	
Facilitation techniques	• Group discussion • Trainer guided • Demonstration • Brainstorming	
Formative assessment methods	• Open ended questions • Matching types questions • Multiple choice questions • Practical exercises	

Integrated/Summative assessment (For specific module)

Integrated situation

The policy of MINEDUC emphasize on hygiene specifically in the kitchens of schools, where all schools are requested to renovate existing kitchens with floor and wall tiles to cope with hygiene issues. In this regards Karongi district has launched a tender for responding to the above said policy and MTK company won the tender of applying tiles in Kitchens of 4 TVET Schools located in KARONGI district, RUBENGERA Sector. As a tiler for a said company, you are requested to apply tiles with following instructions:

- ✓ 10sqm of floor and 10msqm of wall at the same kitchen.
- ✓ All materials are available at the site.
- ✓ Use of spacers;
- ✓ Grouting and cleaning must be respected.

- ✓ The mixing ratio of 1:2 must be respected.
- ✓ Time: 8hours

Resources

Tools	Tape measure, Square, Spades, Horse pipe level, Trowel, Spirit level, Hammer, building line, Spacers, pan, Pincers, Rubber mallet, Electrical drill, Sliding cutter, Marker pen, bucket, chisel, Brush.
Equipment	PPE: - Helmet - Gloves - Goggles - Dust mask - Overall - Safety shoes(boots), Dump level, Mixer, Tile cutter, Wheel barrow
Materials/ Consumables	Water, Oil, Sand, Cement, Grout filler, Tiles, Spacers, building line.

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Prepare the work (20%)	1.1. Appropriate tools are selected according to the work to be performed	Cutting tools, setting tools, batching tools, mixing tools, marking tools, placing tools are selected.			5
	1.2. Equipment is selected according to the work to be performed	PPEs, tile cutter, mortar mixer, wheelbarrow are selected			5
	1.3. Adequate materials are selected according to the standard	Floor & wall tiles, water, cement, sand, spacers are selected.			5
	1.4. Workplace is properly prepared according to the work to be done.	Dust, vegetable, oil and water dirties are not observed.			5
Learning outcome 2: Prepare mortar (20%)	2.1. Mixing place is adequately prepared as per standard requirements.	Mixing place is cleared and cleaned			5
	2.2. Mortar materials are properly collected according to their types.	Water, cement, sand and additives are collected			5
	2.3. Mortar materials are accurately batched in accordance with mix ratio.	Mixing ratio is respected			5
	2.4. Proper mixing of mortar ingredient is well respected.	Mortar mixture is homogeneous			5
Learning outcome 3: Lay tiles (40%)	3.1. Area to be tiled is accurately set out in accordance with designed levels.	The center line is marked			3
		Midpoint of the room is marked			3
		Starting point is allocated			4
	3.2. Tiles are accurately cut	Tiles are accurately measured			3

	according to the place to be used.	Tiles are accurately marked			3
		Tile are accurately cut			4
	3.3. Tiles are properly applied to the surface according to the designed pattern.	Tiles are dressed			4
		Tiles are levelled			4
		The mortar is placed			4
		Tiles are applied to the surface			3
	3.4. Tiles joints are adequately grouted according to the specification.	Joints of tiles are grouted			5
Learning outcome 4: Clean the workplace (20%)	4.1. Tiled surface is properly cleaned after work.	Tiled surface is clean			4
	4.2. Working area is properly cleaned after work.	Working area is clean			4
	4.3. Used tools and equipment are properly cleaned according to their type.	Tools and equipment are clean			6
	4.4. Used tools and equipment are correctly stored according to their category and status.	Tools and equipment are stored			6
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References:

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4. <https://atozbuilding.in/2020/04/tiles-work.html>
5. www.armatile.com